Hallucination Mitigation in Reasoning Models

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Abstract

This project addresses hallucination mitigation in story continuation tasks using the ROCStories dataset. We develop a multi-stage approach combining structured chain-of-thought (CoT) prompting, reward model (RM) reranking, and Direct Preference Optimization (DPO). Starting from a simple baseline (overall quality score: 2.96), we improve to a strict CoT baseline (3.52), then apply RM-based best-of-N reranking (3.50) and DPO fine-tuning (3.96). Our best configuration combines DPO-trained policy with RM reranking, achieving an overall quality score of 4.04—a 36.8% relative improvement over the simple baseline. Results demonstrate that training-time alignment (DPO) and inference-time selection (RM) are complementary strategies for reducing hallucinations while maintaining narrative coherence.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in generating coherent and contextually relevant text. However, they frequently produce hallucinations—content that is factually incorrect, logically inconsistent, or unfaithful to the given context. This phenomenon poses significant challenges for deploying LLMs in applications requiring high reliability and factual accuracy. Our project focuses on mitigating hallucinations in reasoning-oriented generation tasks by improving reward modeling and leveraging preference-based alignment techniques. We specifically target the story continuation task using the ROCStories dataset, where models must generate a single coherent ending sentence given a four-sentence story context. This task is particularly challenging because the model must maintain semantic consistency with established narrative elements, produce commonsense-plausible events, and avoid introducing contradictory or fabricated details that violate the story's internal logic.

1.1 Formal Problem Definition

We define the story continuation task as follows: Given a context sequence c consisting of the first four sentences of a story and a prompt-specific instruction I, the model must generate a continuation y (the fifth sentence).

$$y \sim P_\theta(\cdot|c, I)$$

The objective is to maximize a judge-evaluated quality metric $S(c, y)$, which is a composite score of information completeness, factual accuracy, relevance, logical coherence, and creativity:

$$\theta^* = \underset{\theta}{\operatorname{argmax}}\, \mathbb{E}_{(c)\in\mathcal{D}}[S(c, y)]$$

Subject to the constraint that y must strictly adhere to format constraints $C(I)$ defined by the prompt type (e.g., strict template following).

2 Literature Review

2.1 Hallucination in Large Language Models

Huang et al. (2025) provide a comprehensive survey on hallucination in LLMs, categorizing hallucinations into two main types: factual hallucinations (including contradiction and fabrication) and faithfulness hallucinations (including instruction, context, and logical inconsistencies). The survey attributes these issues to data-related factors such as misinformation and bias, knowledge boundary problems, and limitations in training stages including pre-training, fine-tuning, and RLHF. The authors emphasize that despite LLMs' ability to encode vast factual knowledge, they struggle to recognize their own knowledge limits, leading to confidently stated falsehoods.

Mitigation strategies reviewed include data filtering and model editing during training, fine-tuning and activation steering at inference time, fact-checking, uncertainty estimation, and reasoning-enhancing approaches like chain-of-thought prompting. Retrieval-Augmented Generation (RAG) systems are also discussed for their use of source attribution to improve factual grounding. Our work builds on these insights by combining structured prompting with preference-based alignment.

2.2 Process and Outcome Reward Models

Xie et al. (2025) propose SP-PRM, a dual-consistency framework that improves inference-time alignment in LLMs. Traditional outcome reward models (ORMs) assess complete responses, which leads to inconsistencies when guiding generation step by step. SP-PRM constructs a process reward model (PRM) from an ORM using partial-sequence data, enforcing score consistency (coherent evaluation across partial and complete responses) and preference consistency (aligning partial evaluations with human preferences). Experiments show that SP-PRM consistently outperforms baseline reward-guided search methods, improving GPT-4 evaluation scores by 3.6%–10.3% while reducing hallucinations.

2.3 System 1 to System 2 Reasoning

Zhou et al. (2025) survey the transition from fast heuristic "System-1" behavior to deliberative "System-2" reasoning in LLMs. The survey catalogues CoT variants, inference-time search strategies, and verifier-augmented decoding. Importantly, it contrasts outcome-focused supervision with process-aware learning, arguing that process signals are required to improve faithfulness rather than merely accuracy. The survey standardizes evaluation beyond accuracy to include faithfulness/consistency and step-level validity as first-class metrics. Our approach instantiates several survey-endorsed strategies: structured CoT prompting, joint use of ORM/PRM as reward sources, and inference-time candidate selection.

2.4 Why Language Models Hallucinate

Kalai et al. (2025) argue that hallucinations in language models arise from fundamental statistical pressures rather than flawed data, showing that even with error-free training corpora, the pretraining objective itself leads models to generate errors. They formalize this by reducing generation to an “Is-It-Valid” (IIV) binary classification problem, establishing a direct relationship between generative error rates (including hallucinations) and misclassification rates in supervised learning. Under this view, the same factors that cause error in binary classifiers (such as epistemic uncertainty and unlearnable facts) inevitably manifest as hallucinations in language models. The authors further show that post-training and evaluation exacerbate this issue: most language-model benchmarks mirror standardized human exams and rely on binary metrics like accuracy or pass rate, which reward confident guessing and penalize abstention, thereby incentivizing hallucination. This perspective directly motivates our approach: rather than treating hallucinations as isolated failures, our project targets the reward structure that governs reasoning behavior. By optimizing preference reward models (PRM) and outcome reward models (ORM) within the GRPO framework, and analyzing hallucination onset during multi-step chain-of-thought reasoning, we aim to counteract these misaligned incentives and explicitly discourage fabrication while preserving reasoning depth and coherence.

2.5 Reasoning Models Hallucinate More

Li et al. (2025) show that reinforcement learning–based fine-tuning for reasoning, while improving benchmark performance, systematically increases hallucinations in large reasoning models. They demonstrate that outcome-based RL with binary rewards induces unstable training dynamics: high-variance gradients when correct answers are rare, entropy pressure that encourages exploratory but unfounded generations, and local optima where models converge to confident but incorrect reasoning paths. Importantly, they find that hallucinations primarily arise in intermediate reasoning steps rather than final answers, indicating that standard reward modeling fails to penalize fabrication within chain-of-thought. This work directly informs our project’s focus on mitigating hallucinations through reward design: rather than introducing external verifiers, we study how modifying preference reward models (PRM) and outcome reward models (ORM) within the GRPO framework can similarly reshape incentives during multi-step reasoning. By identifying hallucination onset stages and dynamically balancing these reward components, our approach will suppress hallucinations while still allowing for depth of reasoning.

3 Experimental Design

3.1 Data

We use the ROCStories corpus, which contains five-sentence commonsense stories. Each story consists of four context sentences followed by one ending sentence. We construct our evaluation split by randomly sampling 500 stories from the test set. For reward model training, we construct 3,000 judged samples by combining 2,000 high-quality generations (sampling twice per prompt from GPT-4) with 1,000 lower-quality generations from the base model. We additionally collect 1,000 preference pairs for DPO training.

Split	Stories	Purpose
Test Set	500	Evaluation
RM Training	3,000	Reward Model Training
DPO Training	1,000 pairs	Preference Alignment

Table 1: Dataset statistics and usage. The test set ($N = 500$) is held out exclusively for evaluation. The RM training set consists of 3,000 individually judged completions, while the DPO training set consists of 1,000 paired (winner/loser) samples to align the model with human-like preferences.

3.2 Evaluation Metric

We employ an LLM-as-a-Judge approach using GPT-4o-mini to evaluate continuations. The judge evaluates five dimensions on a 0–5 scale: Information Completeness (IC), Factual Accuracy (FA), Relevance (Rel), Logical Coherence (LC), and Creativity (CE). A summary of the rubric is provided in Table 2. A full rubric is provided in Appendix A.

The final metric, **Overall Quality**, is computed using a quality-gated length formula to penalize "safe but empty" answers while preventing verbosity hacking:

$$S_{quality} = S_{raw} + \lambda \cdot B_{len} - P_{verb}$$

Where:

- S_{raw} is the unweighted mean of the five dimension scores.
- $B_{len} \in [0, 1]$ is a bonus term that rewards outputs within the target range of 40–120 words.
- $\lambda = \left(\frac{\text{IC+REL}}{10}\right)^2$ is a quality gate. This ensures that only answers with high Information Completeness and Relevance benefit from the length bonus.
- P_{verb} is a penalty applied if the length exceeds 120 words without sufficient coherence.

This formula ensures the model is rewarded for substantive, relevant storytelling rather than brevity.

3.3 Simple Baseline

Our simple baseline uses DeepSeek-R1-Distill-Llama-8B (4-bit quantized) with loose prompting that requests only a story continuation without explicit reasoning steps. This represents a "non-rationalized direct decode baseline" where the model generates continuations without chain-of-thought guidance. The simple baseline achieves an Overall Quality score of 2.955, establishing the lower bound for our experiments.

4 Experimental Results

4.1 Baseline

Results. The loose baseline achieves an Overall Quality score of **2.955**. While the model generally stays on topic, it frequently suffers from hallucinated details or generic endings that fail to resolve the specific narrative tension introduced in the context.

4.2 Extension 1: Structured Chain-of-Thought

For our first extension, we hypothesized that enforcing explicit reasoning steps would reduce hallucinations. We experimented with two variations:

- **Moderate**: Provided reasoning guidance with flexible structure.
- **Strict**: Enforced a rigid four-step template (Setup/Reasoning/Check/Answer).

Results. The results revealed a "Prompt Adherence Paradox." The **Moderate** prompt actually underperformed the baseline (2.577), confusing the model into outputting planning text rather than a story. However, the **Strict** template achieved an Overall Quality score of **3.522**, a 19.2% improvement over the baseline. It showed substantial gains in Relevance (+0.50) and Logical Coherence (+0.35). Consequently, we adopted the Strict Prompt as the foundation for all subsequent extensions.

Dimension	1 pt	2 pts	3 pts	4 pts	5 pts
Info Completeness	<20 words; brief conclusion	20-50 words; 1 key point	≥50% key points	≥75% key points	≥90% key points
Factual Accuracy	≥2 core errors	1 core + minor error	≤1 small error	Minor ambiguity only	All facts correct
Relevance	>50% off-topic	30-50% off-topic	≤30% off-topic	<10% redundancy	Fully on-point
Logical Coherence	No chain; answer only	Many jumps; steps missing	Mostly complete; minor leaps	Clear, structured	Exceptionally rigorous
Creativity	Minimal variation	Simple example	Somewhat vivid	Multiple fresh angles	Highly original

Note: A score of 0 indicates complete failure (irrelevant, contradictory, or nonsensical). Decimals (e.g., 3.20, 4.75) are encouraged for fine-grained distinctions.

Table 2: Evaluation rubric summary. The judge assesses five quality dimensions on a continuous 0–5 scale. Higher scores indicate greater faithfulness and coherence, with specific penalties applied for factual contradictions (in Factual Accuracy) and off-topic generation (in Relevance).

Model	**RM (N=3)**			**DPO**
Prompt	Beginning: The boy went to a video arcade. He played his favorite machine. Instruction: Write a continuation of the story starting from the given beginning. Your continuation should describe what happened after the boy played his favorite machine, including any challenges he faced and any interactions with other characters. Format your response as a coherent paragraph continuing the narrative.			Beginning: The boy went to a video arcade. He played his favorite machine. Instruction: Write a continuation of the story starting from the given beginning. Your continuation should...
Completion	The boy had been coming to the arcade for weeks, but today felt different. As he pressed the start button on his favorite machine, the screen flickered before the usual game began, but this time, the pixels distorted, making the characters look warped...	The boy's fingers danced over the buttons, his eyes locked on the screen of his favorite arcade machine—a vintage space shooter that always seemed to draw a crowd. As he reached the final level, the familiar rush of adrenaline kicked in. Sweat beaded on his forehead...	The boy leaned forward, gripping the joystick as his favorite arcade machine flickered to life. He'd played this game a hundred times, but today felt different—his fingers moved with a focused precision he hadn't felt before. The pixelated hero on screen darted through labyrinths...	After playing the machine, the boy felt a strange tingling sensation in his fingertips. He looked around, but the arcade was quiet. He tried to shake it off, but the tingling grew stronger. Then, he noticed a flickering light above the machines...
Score	4.6 ✅	3.6	2.8	5.0

Figure 1: Process comparison between extensions. The table illustrates the input-output flow for each stage. For Reward Model Reranking (Extension 2), we generate $N = 3$ candidates; the green checkmark indicates the candidate selected for having the highest predicted reward score. In contrast, DPO (Extension 3) directly aligns the generation policy, producing high-quality outputs without requiring multiple samples.

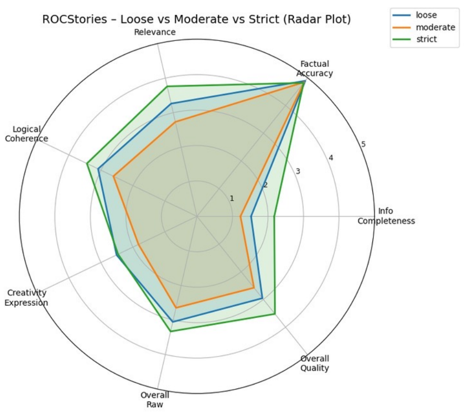

Figure 2: Radar plot comparing evaluation dimensions across prompt types (Loose, Moderate, Strict). Strict prompting shows expanded coverage across most dimensions. Factual Accuracy remains consistently high (∼5) across all conditions, while Information Completeness shows the largest variation.

4.3 Extension 2: Reward Model Reranking

Our second extension integrates a Reward Model (RM) into the inference pipeline to improve story continuation quality through best-of-N reranking. The RM learns an explicit scoring function $r(x, y)$ that approximates GPT-4.1-mini's rubric-based evaluation.

Training. We construct 3,000 training samples: 2,000 GPT-4-generated continuations (querying twice per prompt for 1,000 stories) plus 1,000 lower-scoring outputs from the base DeepSeek model. Each sample pairs the story beginning with a continuation and its GPT-4.1-mini rubric scores across five dimensions. The RM uses a regression head on Llama-8B with 4-bit quantization and LoRA fine-tuning (batch_size=1, gradient_accumulation=32, epochs=1), predicting `overall_raw` (mean of five dimension scores) as the target.

Inference. At inference time, we generate $N = 3$ candidate continuations using the strict prompt and select the highest-scoring candidate according to the RM.

Results. RM reranking with the strict prompt achieves an Overall Quality score of **3.503**. While this shows a marginal decline from the strict baseline (-0.019), the RM demonstrates effective filtering with improved Relevance (+0.21) despite reduced Creativity (-0.37). The creativity reduction suggests the RM favors "safer" outputs, a trade-off we address in our next extension.

4.4 Extension 3: Direct Preference Optimization

Our third extension applies Direct Preference Optimization (DPO) to directly shape the policy distribution from pairwise preference data, eliminating inference-time sampling overhead.

Training Data. We construct 1,000 preference pairs (x, y^+, y^-) from GPT-4.1-mini judgments. For each story beginning x, we pair a high-scoring GPT-4-generated continuation (y^+) with a lower-scoring base model output (y^-). Pairs are selected to have clear quality separation based on `overall_raw` scores.

Training. DPO fine-tuning is applied with LoRA to the base DeepSeek model. Unlike RM-based approaches, DPO optimizes the policy directly without requiring a separate reward model at inference time, making single-sample generation more efficient.

Results. DPO achieves a substantial improvement, reaching an Overall Quality score of **3.961**—a 12.5% gain over the strict CoT baseline. Notably, DPO improves across all dimensions simultaneously: Information Completeness (+0.48), Relevance (+0.45), Logical Coherence (+0.47), and Creativity (+0.66). The creativity improvement is particularly striking given that RM reranking alone *reduced* creativity, suggesting that DPO learns to generate higher-quality outputs rather than simply filtering for safety. A comprehensive breakdown of performance across all stages is presented in Figure 1, highlighting the recovery of creativity scores in the final DPO + RM configuration.

Method	IC	FA	REL	LC	CE	OQ
Loose Baseline	1.524	4.893	3.264	3.090	2.511	2.955
Moderate CoT	1.226	4.844	2.734	2.611	1.826	2.577
Strict CoT	2.177	4.819	3.763	3.439	2.464	3.522
Strict + RM	2.270	4.549	3.969	3.341	2.091	3.503
Strict + DPO	2.655	4.698	4.208	3.906	3.125	3.961
DPO + RM	**2.823**	**4.765**	**4.311**	**4.018**	**3.192**	**4.044**

Table 3: Main evaluation results ($N = 500$). The combined DPO + RM approach achieves the highest performance.

Note: **IC**: Info Completeness, **FA**: Factual Accuracy, **REL**: Relevance, **LC**: Logical Coherence, **CE**: Creativity, **OQ**: Overall Quality.

4.5 Final Configuration: DPO + Reward Model

Finally, we test whether training-time alignment (DPO) and inference-time selection (RM) are complementary by combining both approaches. The DPO-trained policy generates $N = 3$ candidates, and the RM selects the highest-scoring output.

Results. The combined approach achieves an Overall Quality score of **4.044**—the highest across all configurations. This represents a **36.8% relative improvement** over the simple baseline and a 14.8% gain over the strict CoT baseline. The combination outperforms DPO alone (+0.08) and substantially outperforms RM alone (+0.54), confirming that the two methods provide complementary benefits: DPO improves the quality of generated candidates, enabling the RM to select from a stronger pool.

4.6 Visual Analysis

To better understand the performance gains, we visualize the results across three dimensions: distribution, progression, and interaction.

Progressive Improvement. Figure 3 illustrates the stepwise improvement from the baseline to our final model. The DPO stage provides the steepest ascent in performance, particularly in recovering the Creativity scores that were penalized by the Reward Model in the previous stage.

DPO and RM Interaction. As shown in Figure 4, the interaction between training-time alignment (DPO) and inference-time search (RM) is additive.

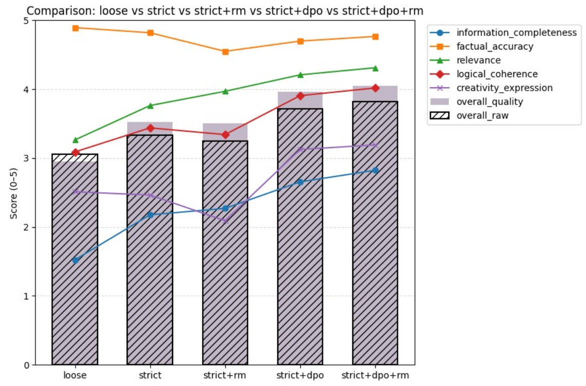

Figure 3: Progressive improvement across methods. Bar heights represent overall quality (solid) and overall raw (hatched). Lines show individual dimension trends. DPO provides the largest single improvement, with RM reranking adding incremental gains when combined.

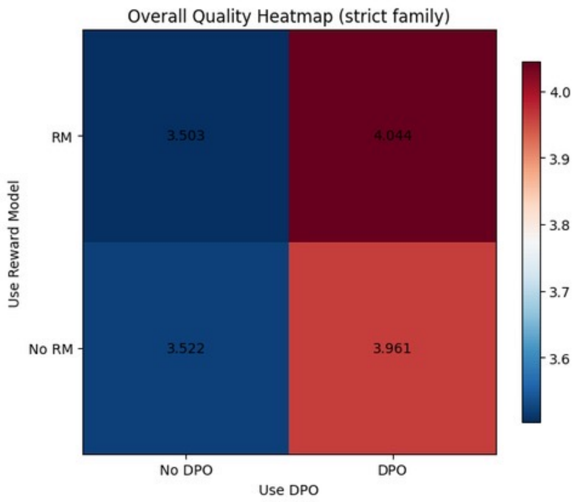

Figure 4: Interaction between DPO and RM within the strict prompt family. DPO provides the primary improvement (3.52→3.96), while RM adds further gains (3.96→4.04).

While DPO contributes the majority of the gain (improving the strict baseline from 3.52 to 3.96), the Reward Model is able to squeeze out further gains (to 4.04) by selecting the best samples from the improved policy.

Distribution Shift. Finally, Figure 5 demonstrates the distributional shift in quality scores. The Loose Baseline (blue) is centered near 3.0, whereas the DPO + RM configuration (orange) shifts the mass significantly to the right, with a peak density near 4.5. This confirms that our method not only increases the average score but consistently reduces the frequency of low-quality (hallucinated) outputs.

5 Error Analysis and Limitations

While our best model (Strict + DPO + RM) achieves substantial improvements over baselines, we identify several categories of remaining errors and limitations that warrant further investigation.

Hallucination Categories. We manually analyzed 50 randomly sampled outputs from our best model and categorized errors according to our Factual Accuracy rubric:

- **Implausible escalation (∼12%)**: The model occasionally introduces events that, while not directly contradicting the context, represent unlikely jumps in narrative scope (e.g., a character learning to cook suddenly entering a competition). These cases receive Factual Accuracy scores of 2–3 under our rubric for introducing "major unsupported events."
- **Entity fabrication (∼8%)**: Introduction of new characters, objects, or locations not established in the four-sentence context. Per our guidelines, such cases incur at least a 1.0 point deduction in Factual Accuracy.
- **Temporal inconsistencies (∼6%)**: Misalignment with the implied timeline, particularly in stories involving sequences of events or time-sensitive situations.

Trade-offs Between Dimensions. Our results reveal inherent tensions between evaluation dimensions:

- **Creativity vs. Safety**: RM reranking alone reduces Creativity by 0.37 points compared to the strict baseline, as the reward model favors "safer" outputs with fewer novel elements. The DPO + RM combination partially mitigates this (CE=3.19 vs. 2.09 for RM-only) by generating higher-quality candidates for selection.
- **Information Completeness vs. Factual Accuracy**: Outputs that attempt to provide more complete narrative closure (higher IC) risk introducing unsupported details (lower FA). Our quality-gated length bonus attempts to balance this, but the tension remains.

Methodological Limitations.

- **LLM-as-a-Judge reliability**: Our evaluation relies on GPT-4.1-mini as a proxy for human judgment. While we employ detailed rubrics

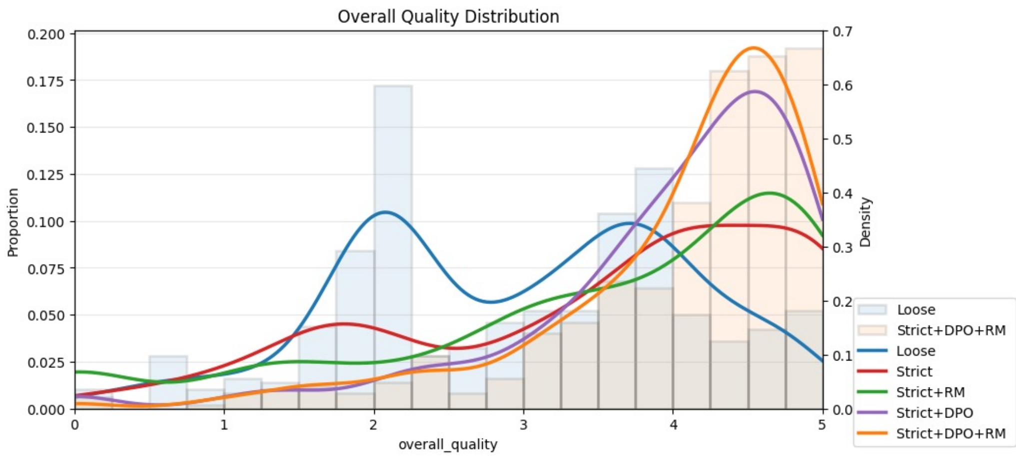

Figure 5: Distribution of Overall Quality scores across methods. The DPO + RM combination (orange) shows a clear rightward shift toward higher quality scores compared to the loose baseline (blue histogram). Kernel density estimates show DPO + RM achieving peak density near 4.5.

and calibration rules, the judge may exhibit biases not captured in our analysis. Inter-annotator agreement with human evaluators was not measured.

- **Model scale constraints**: Our experiments use DeepSeek-R1-Distill-Llama-8B with 4-bit quantization due to computational limitations. Larger models or full-precision inference may yield different trade-offs between methods.
- **Domain specificity**: Results are specific to the ROCStories dataset (short, commonsense narratives). Generalization to longer-form generation, technical domains, or multi-turn dialogue remains untested.
- **Reward model coverage**: The RM was trained on 3,000 samples from a limited distribution. Out-of-distribution prompts may receive unreliable scores, potentially causing the best-of-N selection to fail silently.

Future Directions. These limitations suggest several avenues for improvement: (1) incorporating process reward models (PRMs) to catch step-level hallucinations before they propagate; (2) human evaluation studies to validate LLM-as-a-Judge reliability; (3) scaling experiments to test whether larger models exhibit similar DPO + RM complementarity; and (4) domain transfer experiments to assess generalization.

6 Conclusions

This project demonstrates that combining training-time alignment with inference-time selection effectively reduces hallucinations in narrative generation. Our key findings are:

1. **Structured prompting establishes a strong foundation.** The strict CoT template alone improves overall quality by 19.2% over naive generation, validating that explicit reasoning constraints benefit coherent story generation. However, partial structure (moderate prompting) can be counterproductive, underperforming even the loose baseline.
2. **DPO provides the largest single improvement.** Direct preference optimization boosts performance by 12.5% over the strong baseline, improving all evaluation dimensions simultaneously. Unlike RM reranking, DPO increases creativity (+0.67) rather than suppressing it, suggesting it learns to generate higher-quality outputs rather than simply filtering for safety.
3. **RM and DPO are complementary.** While RM alone shows marginal gains over the strict baseline, combining RM reranking with the DPO-trained policy achieves our best result (4.044). This confirms that better candidates enable more effective selection—the two methods leverage complementary strengths.
4. **Total improvement of 36.8%.** From simple baseline (2.955) to DPO + RM (4.044) represents substantial progress in reducing hallucinations while maintaining narrative quality.

Our approach does not reach state-of-the-art performance on ROCStories benchmarks, primarily due to model scale constraints (8B parameters with 4-bit quantization). However, our results demonstrate that alignment techniques can significantly improve smaller models' reliability in creative generation tasks. Future work could explore process reward models (PRMs) for step-level supervision, scaling experiments to test generalization of DPO + RM complementarity, and human evaluation studies to validate LLM-as-a-Judge reliability.

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A Judge System Prompt

The following system prompt is provided to GPT-4o-mini for evaluation.

Component	Instruction Content
Role & Context	You are an expert evaluator for narrative story continuations. You will be given: • Beginning: the initial context for the story. • Model continuation: the candidate continuation. Judge ONLY the model continuation.
Task Definition	This is a story continuation faithfulness task. The continuation should be semantically consistent with the context and should not introduce hallucinated core entities or events unsupported by the beginning.
Scoring Scale	Score each dimension from 0 to 5. Use decimals with two digits when helpful (e.g., 3.20, 4.75). A score of 5 should be rare and reserved for truly exceptional cases.
1. Information Completeness	Measures whether the continuation meaningfully completes the narrative implied by the context. • 0: empty, nonsensical, or unrelated continuation. • 1: extremely thin ending with minimal closure. • 2: partial ending that addresses only one small aspect. • 3: reasonably completes the main thread but leaves key outcomes vague. • 4: well-rounded ending resolving most salient issues. • 5: exceptionally complete and satisfying resolution.
2. Factual Accuracy & Context Faithfulness	Measures consistency with context entities, events, and causal state. • 0: multiple severe contradictions or impossible events. • 1: clear contradiction or major impossible claim. • 2: major unsupported event or entity altering story meaning. • 3: unsupported detail changing interpretation. • 4: minor unverifiable detail without core impact. • 5: fully consistent with context and commonsense. Guideline: Deduct at least 1.0 point for unsupported core entities or events.
3. Relevance & Constraint Following	Measures how tightly the continuation stays on-topic and follows prompt intent. • 0: completely off-topic or ignores constraints. • 1: largely off-topic with heavy violations. • 2: significant drift with unnecessary elements. • 3: mostly relevant with some fluff. • 4: strongly on-topic with minor extras. • 5: fully on-topic and strictly constraint-following.
4. Logical Coherence & Clarity	Measures stepwise narrative logic and clarity. • 0: incoherent or self-contradictory. • 1: almost no narrative reasoning. • 2: multiple logical jumps. • 3: generally coherent with some unclear transitions. • 4: clear and plausible progression. • 5: exceptionally clear and well-structured.
5. Creativity & Expression	Measures originality and engaging writing while remaining faithful to context. • 0: dull or template-like. • 1: very plain and repetitive. • 2: generic with few details. • 3: some vivid or interesting detail. • 4: fluent and engaging style. • 5: highly original and memorable.

Table 4: Judge system prompt and scoring rubric.

Component	Instruction Content
Calibration Rules	• Do not default to 5s for well-formed text. • Use 3.5 to 4.6 for good but not outstanding continuations. • Reflect small differences using decimals when appropriate. • Avoid identical scores across many samples unless justified. • Think carefully but do not output reasoning.

Output Format	Return ONLY a JSON object:
	`{`
	`"information_completeness": 0-5,`
	`"factual_accuracy": 0-5,`
	`"relevance": 0-5,`
	`"logical_coherence": 0-5,`
	`"creativity_expression": 0-5,`
	`"comments": "short comment (<=30 words)"`
	`}`

Table 5: Judge calibration rules and output format.

The full prompt includes detailed rubrics for each dimension with calibration rules to ensure consistent scoring. The judge returns a JSON object with scores and a brief comment ($\leq$ 30 words).